

A Generalized Algorithm for Multi-Objective Reinforcement Learning

Summary

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1 Problem Setup

Multi-objective reinforcement learning (MORL) deals with problems where agents must optimize multiple competing objectives simultaneously. Traditional approaches often require specifying preferences in advance or use scalarization methods that may miss important parts of the solution space. This paper introduces a generalized MORL algorithm that can learn the entire Pareto frontier and adapt to new preferences without retraining.

2 Thesis

The central thesis is that MORL can be generalized to learn the complete set of optimal policies across all possible preferences, enabling efficient adaptation to new preferences without additional training. The authors argue that by learning the entire control frontier rather than just single policies, agents can achieve better performance and flexibility in multi-objective environments.

3 Key Concepts

Control Frontier: The set of all optimal policies that achieve the best possible trade-offs between different objectives, representing the Pareto frontier in policy space.

Envelope Method: An approach that maintains a set of policies covering the entire control frontier, allowing for adaptation to any preference without retraining.

Scalarization: The traditional approach of combining multiple objectives into a single scalar reward using weighted sums, which can miss parts of the Pareto frontier.

Policy-Preference Alignment: The mapping between user preferences (weights on objectives) and optimal policies on the control frontier.

Adaptive MORL: The ability to adapt to new preferences by selecting appropriate policies from the learned control frontier without additional training.

4 Methods

The authors develop a generalized MORL framework with several key components:

1. **Envelope Learning:** Instead of learning single policies, the algorithm learns an envelope of policies that covers the entire control frontier.

2. **Preference Inference:** The system can infer user preferences from observed behavior and select appropriate policies from the learned frontier.
3. **Frontier Approximation:** Methods for efficiently representing and approximating the high-dimensional control frontier using a finite set of policies.
4. **Adaptive Selection:** Mechanisms for selecting the best policy from the learned frontier given new user preferences.
5. **Generalized Updates:** Learning rules that maintain coverage of the entire frontier during training rather than converging to single policies.

The key innovation is the shift from learning individual policies to learning the complete set of optimal policies across all preferences.

5 Results

The authors demonstrate their approach on several multi-objective domains:

- **Resource Management:** In a resource collection environment, the envelope method achieves better performance across all preference combinations compared to scalarization approaches.
- **Dialog Systems:** In task-oriented dialog policy learning, the envelope method achieves higher success rates and better adaptation quality than single-objective baselines.
- **Game Playing:** In a multi-objective SuperMario environment, the envelope method outperforms scalarized approaches across different preference combinations.
- **Preference Adaptation:** The system can quickly adapt to new preferences by selecting appropriate policies from the learned frontier without retraining.
- **Coverage Analysis:** Experiments show that the envelope method provides better coverage of the true Pareto frontier compared to scalarization approaches.
- **Efficiency:** While learning the full frontier requires more computation initially, adaptation to new preferences is much more efficient than retraining.

The results demonstrate that learning the complete control frontier enables better performance and more flexible adaptation than traditional approaches.

6 Significance

This work represents an important advance in multi-objective reinforcement learning:

- **Generalization:** Provides a truly generalized approach to MORL that can handle any preference without retraining, overcoming limitations of traditional scalarization methods.
- **Practical Impact:** Makes MORL more practical for real-world applications where preferences may change or be unknown in advance.
- **Theoretical Foundation:** Establishes a theoretical framework for understanding and approximating control frontiers in high-dimensional objective spaces.

- **Adaptability:** Enables agents to adapt to changing user preferences or multiple stakeholders with different priorities.
- **Coverage:** Ensures that no part of the Pareto frontier is missed, addressing a fundamental limitation of scalarization approaches.

The paper provides a foundation for more flexible and practical multi-objective reinforcement learning systems that can handle the complexity of real-world decision-making with multiple competing objectives.